CHAPTER 3

SYSTEMS OF LINEAR ALGEBRAIC EQUATIONS AND THEIR SOLUTIONS

In many engineering problems there is a constant need for solving systems of linear equations. It could be said with full responsibility that solving of such equations constitutes one of the most common and important problems of the numerical mathematics. The system of n linear equations can be written in the following expanded form:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

Using the definitions (notions) of the square matrix and the column matrix (vector), the system of n linear equations can be represented by the following equivalent matrix equation:

$$\mathbf{A} \,.\, \mathbf{X} = \mathbf{B}$$

Where:

$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$ is the square matrix of coefficients

$\mathbf{B} = [b_1 \quad b_2 \quad \cdots \quad b_n]^T$ is the vector of free terms

$\mathbf{X} = [x_1 \quad x_2 \quad \cdots \quad x_n]^T$ is the vector of variables

The transposition symbol “T” is used for the vectors **B** and **X**, which are in fact column matrices. Solution of the system equations consists in finding such values for every component of the vector of unknowns **X** that all system of equations are simultaneously satisfied. This assertion is legitimate only when it is assumed that such solution exists.

In the opposite case, the whole effort, undertaken in order to determine such solution, would be in vain. In order to avoid such undesirable conditions, we should investigate in advance the existence of a unique nontrivial solution – the task for which the analysis of the square coefficient matrix **A** and calculation of its determinant can help. The fundamental forms of square matrices and the formula used for calculating their determinants are given below for the particular case of the third-order square matrix (n = 3).

Symmetric matrix:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 2 \\ -1 & 2 & 4 \end{bmatrix}, \quad (a_{ij} = a_{ji})$$

Upper triangular matrix:

$$\mathbf{U} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Lower triangular matrix:

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$

Diagonal unitary matrix:

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Zero matrix:

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The variable D (det A) defined below is called the determinant of the square matrix A of order n:

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum (-1)^k a_{1\alpha} a_{2\beta} \ldots a_{n\omega}$$

Where the indexes α, β . . . ω denote all among the n! Possible permutations of the numbers 1, 2, 3 … n, and k is the number of inversions in a particular permutation.

According to this definition, the determinant of the second-order matrix (n = 2) is:

$$\det \boldsymbol{A} = a_{11}a_{22} - a_{12}a_{21}$$

In case of the third-order matrix (n = 3), we have:

$$\det \boldsymbol{A} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{21}a_{32}a_{13} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{33}a_{12}a_{21}$$

In the general case (n>3), the calculation of the determinant of the square matrix is a cumbersome/bulky task. Therefore, as a rule, we use an indirect method based on the properties of the triangular matrix **T** (upper or lower), having the determinant equal to:

$$\det \boldsymbol{T} = t_{11}.t_{22}.t_{33}.\cdots.t_{kk}.\cdots.t_{nn}$$

Where t_{kk} is thek^{th} element of the main diagonal of this matrix.

Another property that can also be used for this purpose is the equality of determinants of a square matrix **A** and the equivalent triangular matrix **T**, if only the rows and columns are not permuted in the process of the transformation of the matrix **A** into the matrix **T**. This transformation can be made by eliminating the unknowns, i.e., in the same way as in case of the elimination process described step by step in Gauss Elimination method below. According to the formula the above formula, we have det **E** = 1 and det **0** = 0. The necessary and sufficient condition for the existence of a solution of the equation system is that the determinant D of the coefficient matrix A is distinct from zero. The matrix for which this condition is satisfied is called nonsingular. When D = 0, the equation system under consideration can have either no solution or an infinite number of solutions. This property has the following simple geometrical interpretation in the case n = 3. Each equation of the system describes a plane in the three-dimensional space. The intersection of the two planes P1 and P2, shown in Figure below, represents the straight line that intersects the third plane P3 at point S. The coordinates (x_{1s}, x_{2s}, x_{3s}) of this point represent the solution being sought.

In case when D = 0, some of these planes are parallel or identical.

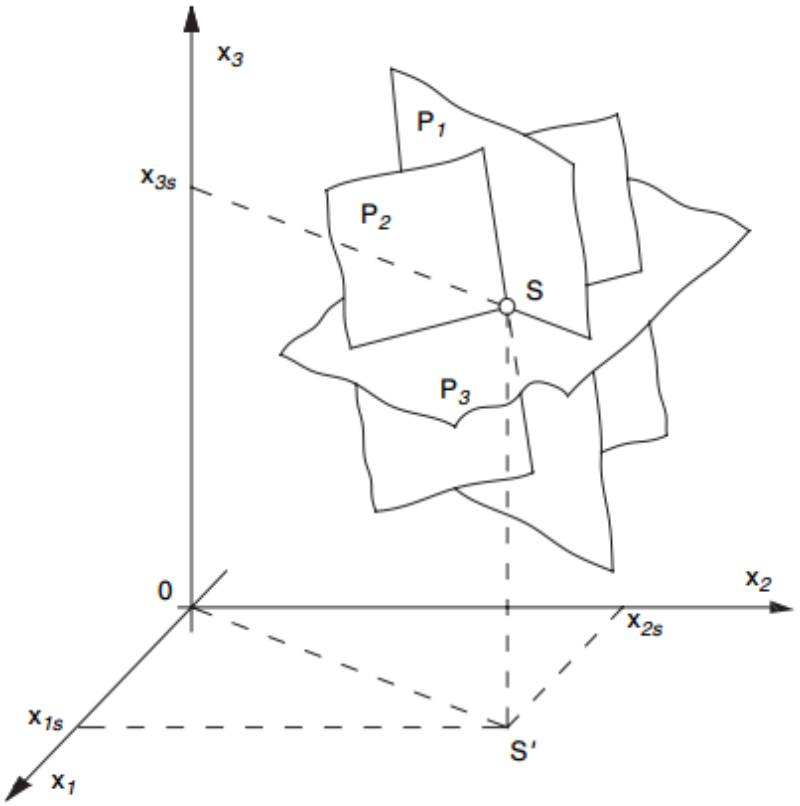


Fig.

The methods used for numerical solution of the systems of linear equations can be classified as follows:

- direct (simple) methods
- iteration methods

In case of the direct methods, explicit recursive formulas are used to determine the components of the vector X constituting the solution, and it is not necessary to know the initial approximate solution (starting point). A classical example of such direct method is the Cramer's rule explained below for solving a system of two equations:

$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

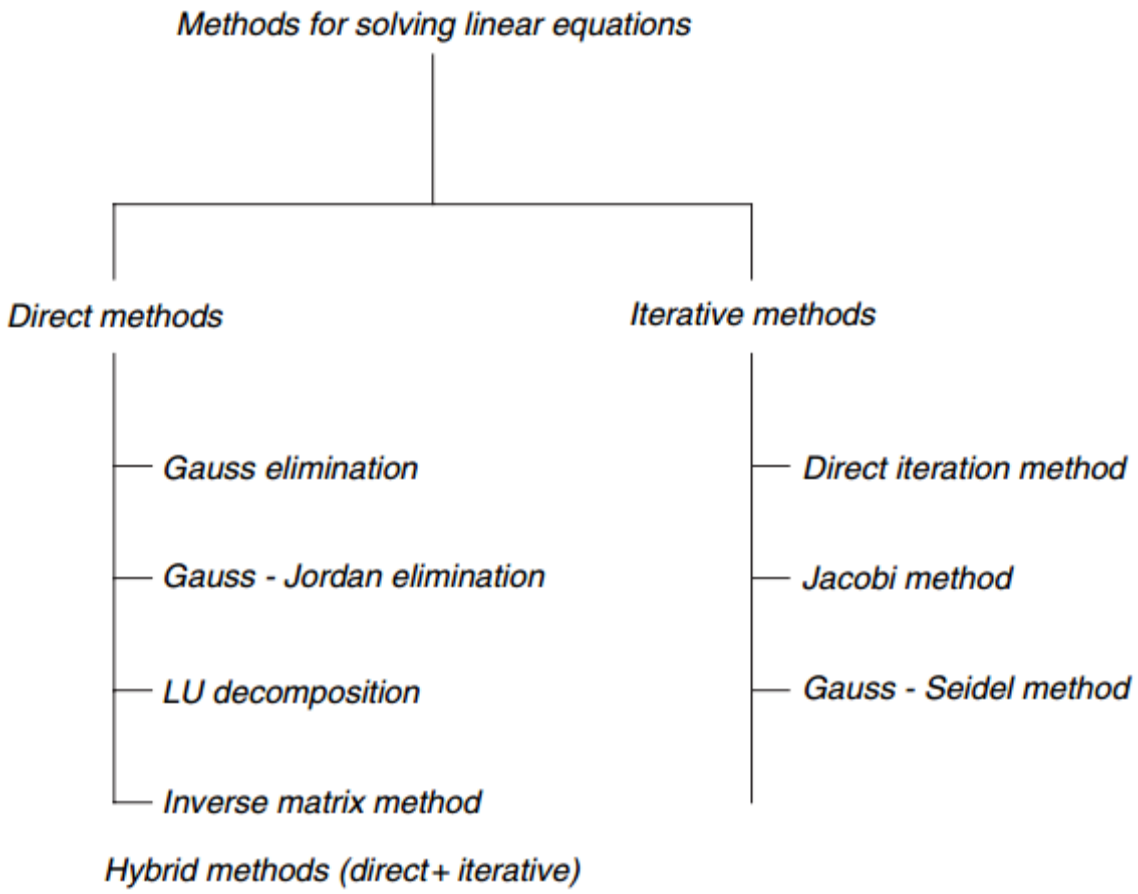
$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$D_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix} = b_1a_{22} - b_2a_{12}$$

$$D_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix} = a_{11}b_2 - a_{21}b_1$$

$$x_1 = D_1/D \qquad x_2 = D_2/D \qquad when\ D \neq 0$$

In case of larger equation systems (n > 2), this rule is numerically ineffective and hence is of little practical use. The main advantages of direct methods are their simplicity and universality. The most important disadvantages are the necessity to store (in the computer memory) the whole coefficient matrix **A** during the computing process and the effect of computing error accumulation, which is specially inconvenient in case of very large equation systems, such as for n>100. The effect of computing error accumulation is absent when we use the iteration methods often called the consecutive iterations methods. They are mainly used for solving the large equation systems. Unfortunately, the knowledge of an approximate solution, ensuring convergence of the computation process, is necessary to start the solving procedure. The basic methods belonging to both groups will be discussed in this chapter, see Figure below.



Direct Methods

1) The Gauss Elimination Method

In order to explain the main concept of the Gauss elimination method, let us consider the following system of three equations in three unknowns:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

In order to eliminate the unknown variable x_1 from the second equation, we add the first equation to the second equation multiplied by$-\ a_{21}/a_{11}$. In a similar way, we

multiply the first equation by$-a_{31}/a_{11}$ and we add it to the third equation. Now we have the following equation system:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2$$

$$a'_{32}x_2 + a'_{33}x_3 = b'_3$$

Where:

$a'_{22} = a_{22} - a_{12}a_{21}/a_{11}\,,\ a'_{23} = a_{23} - a_{13}a_{21}/a_{11}\,,\ b'_2 = b_2 - b_1a_{21}/a_{11},$

$a'_{32} = a_{32} - a_{12}a_{31}/a_{11}\,,\ a'_{33} = a_{33} - a_{13}a_{31}/a_{11}\,,\ b'_3 = b_3 - b_1a_{31}/a_{11},$

Subsequently, we can eliminate the variable x_2 from the third equation of the system above. For this end, we multiply the second equation by the term $-a'_{32}/a'_{22}$ and add it to the third equation of the system. Finally, we obtain the system:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2$$

$$a''_{33}x_3 = b''_3$$

Where:

$$a''_{33} = a'_{33} - a'_{23}a'_{32}/a'_{22} \qquad and \qquad b''_3 = b'_3 - b'_2a'_{32}/a'_{22}$$

This transformed equation system is mathematically completely equivalent to the original system equation. The coefficient matrix of this system has the triangular form, which means that the first stage of the procedure, called the elimination stage or, more colloquially, the upward movement, is completed. According to the formula

for the determinant of upper or lower triangular matrix, the determinant of this matrix is equal to:

$$D = a_{11}.a'_{22}.a''_{33}$$

And has the same value as the determinant of the coefficient matrix **A** of the original system equation. In case this determinant is different from zero, the transition to the second stage of the procedure is legitimate.

This second stage is called the substitution stage, or the backward movement, and begins with the determination of x_3 from the third equation and substitution of the value of x_3 obtained in this way in the second equation. After this substitution, the second equation of the system contains only one unknown variable, namely x_2, which can be found in the elementary way. The values of x_3 and x_2 calculated in this way are then substituted in the first equation of the system, which reduces to the one-variable equation in x_1, which can be solved in an equally simple way.

The second and the third equation of the system:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a'_{22}x_2 + a'_{23}x_3 = b'_2$$

$$a'_{32}x_2 + a'_{33}x_3 = b'_3$$

Can be interchanged in order to obtain the condition in which the coefficients in the main diagonal have the maximum absolute values. In this manner, the numerical error of the method is reduced. Additionally, the interruption of the count after the incidental occurrence of the division by zero becomes impossible. The method improved in this way is called the **Gauss elimination method** with the choice of the main (pivotal) element, called the pivoting strategy. For the arbitrary n ≥ 3, the

variable elimination process used in the Gauss method (stage 1) can be considered as determination of the matrix series:

$$\mathbf{A}^{(1)} = \mathbf{A},\ \mathbf{A}^{(2)},\ \mathbf{A}^{(3)},\ \ \cdots,\ \mathbf{A}^{(i)},\ \cdots,\ \mathbf{A}^{(n)};$$

$$\mathbf{B}^{(1)} = \mathbf{B},\ \mathbf{B}^{(2)},\ \mathbf{B}^{(3)},\ \ \cdots,\ \mathbf{B}^{(i)},\ \cdots,\ \mathbf{B}^{(n)}$$

Where the matrices $\mathbf{A}^{(i)}$ and $\mathbf{B}^{(i)}$ have the form:

$$\mathbf{A}^{(i)} = \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1i}^{(1)} & \cdots & a_{1n}^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2i}^{(2)} & \cdots & a_{2n}^{(2)} \\ \vdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & a_{ii}^{(i)} & \cdots & a_{in}^{(i)} \\ \vdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ 0 & 0 & & a_{ni}^{(i)} & & a_{nn}^{(i)} \end{bmatrix} \qquad \mathbf{B}^{(i)} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_i^{(i)} \\ \vdots \\ b_n^{(i)} \end{bmatrix}$$

According to the procedure of elimination of the consecutive unknown variables described above, the elements of these matrices are determined by using the following expressions:

$$a_{jk}^{(i+1)} = a_{jk}^{(i)} - \frac{a_{ji}^{(i)}}{a_{ii}^{(i)}} * a_{ik}^{(i)} \qquad\qquad b_j^{(i+1)} = b_j^{(i)} - \frac{a_{ji}^{(i)}}{a_{ii}^{(i)}} * b_i^{(i)}$$

Where:

$i = 1, 2, 3, \ldots, n; \qquad j = i+1, i+2, i+3, \ldots, n; and\ k = i+1, i+2, i+3, \ldots, n$

In a case when $a_{ii}^{(i)} \approx 0$ dividing by zero may occur, see formula above. Such computational menace can be eliminated by an appropriate choice of the pivotal element, also called pivot. This protection consists in the choice, among elements a $a_{ii}^{(i)}, a_{i+1.i}^{(i)}, a_{i+2.i}^{(i)}, \ldots, a_{ni}^{(i)}$ belonging to the i – column, of the element having the

largest nonzero absolute value. Let the row k – of the matrix $\mathbf{A}^{(i)}$ be the row, for which:

$$\left|a_{ki}^{(i)}\right| = max\left|a_{ji}^{(i)}\right|, \qquad i \le j \le n$$

Then, the row k of the matrix $\mathbf{A}^{(i)}$ should be interchanged with the ith row. At the same time, the elements $b_i^{(i)}$ and $b_k^{(i)}$ of the column matrix (vector) $\mathbf{B}^{(i)}$ should be interchanged. In a similar way, the process of calculating the matrix $\mathbf{A}^{(i)}$ and $\mathbf{B}^{(i)}$ should be continued up to the position $i = n$. The resulting equation system, equivalent to the original system $\mathbf{A} \,.\, \mathbf{X} = \mathbf{B}$, is:

$$\mathbf{A}^{(n)} . \mathbf{X} = \mathbf{B}^{(n)}$$

In which the coefficient matrix $\mathbf{A}^{(n)}$ is the upper triangular matrix. In the process of finding the pivot in the ith iteration, if we obtain:

$$\left|a_{ki}^{(i)}\right| = max\left|a_{ji}^{(i)}\right| \le \varepsilon$$

Where ε is a given, sufficiently small positive number (e.g., $\varepsilon = 10^{-16}$), the whole process should be interrupted, because it means that the determinant D of the matrix of coefficients ($\mathbf{A}, \mathbf{A}^{(n)}$) is equal to zero. In the opposite case ($D \neq 0$), we should pass to the next substitution stage. The essential meaning of this uncomplicated stage was explained above using an example of the three equations in three variables. When n ≥ 3, the particular terms of the desired solution can be found by using the following recursive formula:

$$x_i = \frac{1}{a_{ii}^{(n)}}\left[b_i^{(n)} - \sum_{j=i+1}^{n} a_{ij}^{(n)} . x_j\right]$$

Where: $i = n, n-1, n-2, \ldots, 1$.

Example: Let us solve the system of three linear equations by using the Gauss elimination method:

$$10x_1 - x_2 + 0x_3 = 7$$

$$-3x_1 + 2x_2 + 6x_3 = 4$$

$$5x_1 - x_2 + 5x_3 = 6$$

In the first step, we eliminate the variable x_1 from the second and the third equation. For this purpose, we multiply both sides of the first equation by the number 0.3 and add it to the second equation. In a similar way, we multiply the first equation by -0.5 and add it to the third equation. As a result of these operations, we obtain:

$$10x_1 - x_2 + 0x_3 = 7$$

$$-0.1x_2 + 6x_3 = 6.1$$

$$2.5x_2 + 5x_3 = 2.5$$

In order to reduce the rounding error, the second and the third equation should be interchanged.

$$10x_1 - x_2 + 0x_3 = 7$$

$$2.5x_2 + 5x_3 = 2.5$$

$$-0.1x_2 + 6x_3 = 6.1$$

Our next step is the elimination of the variable x_2 from third equation. For this end, the second equation should be multiplied by 1/25 and then added to the third equation. After this operation, we obtain the system:

$$10x_1 - x_2 + 0x_3 = 7$$

$$2.5x_2 + 5x_3 = 2.5$$

$$6.2x_3 = 6.2$$

The coefficient matrix of this transformed equation system has the form of a triangular matrix. The determinant of this matrix, calculated using the formula given earlier as:

$$\mathrm{D} = 10 * 2.5 * 6.2 = 155$$

The nonzero value of this determinant shows that the equation system under consideration has one nontrivial solution. It follows directly from the third equation that $x_3 = 1$. After substituting the value $x_3 = 1$ in the second equation, we obtain:

$$2.5x_2 + 5 * 1 = 2.5 \quad \leftrightarrow \quad x_2 = -1$$

Substituting $x_3 = 1$ and $x_2 = -1$ in the first equation, we obtain $x_1 = 3/5$. Finally, our complete solution is as follows: $x_1 = 3/5\,, x_2 = -1\ and\ x_3 = 1$. The formulas and the procedure for choosing the pivoting element, explained in this section, served as a base for the design of the computer program GAUSS. This program was subsequently used to solve the following system of six linear equations (n = 6):

$$\begin{bmatrix} 6 & -3 & 2 & 1 & -1 & 1 \\ -3 & -7 & 0 & 4 & -2 & 1 \\ 4 & -3 & 6 & -1 & 2 & 1 \\ 2 & 4 & 5 & -7 & -3 & 2 \\ -1 & 5 & -4 & 0 & 8 & -2 \\ 3 & 0 & 4 & -2 & 5 & -6 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \\ 28 \\ -6 \\ 25 \\ -4 \end{bmatrix}$$

During the elimination process, accompanied by an operation of choosing the pivots, this equation system is being transformed to the following equivalent form:

$$\begin{bmatrix} 6 & -3 & 2 & 1 & -1 & 1 \\ 0 & -8.5 & 1 & 4.5 & -2.5 & 1.5 \\ 0 & 0 & 4.921569 & -4.686275 & -4.137255 & 2.549020 \\ 0 & 0 & 0 & 2.135458 & 6.784861 & -2.199203 \\ 0 & 0 & 0 & 0 & 5.264925 & 0.1343285 \\ 0 & 0 & 0 & 0 & 0 & -6.612686 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}$$

$$= \begin{bmatrix} 11 \\ -0.5 \\ -9.372549 \\ 29.270920 \\ 27.1306 \\ -39.67612 \end{bmatrix}$$

Solution of this equation system, determined during the second, substitution stage, is the vector $X \equiv (x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5, x_6 = 6)$.

The Gauss–Jordan Elimination Method

In order to explain the essence of the Gauss–Jordan method, let us reconsider the system of three linear equations in three unknowns $I_1, I_2\ and\ I_3$ interpreted here as the complex amplitudes of the currents in a three-mesh electrical circuit.

$$R_{11}\boldsymbol{I}_1 + R_{12}\boldsymbol{I}_2 + R_{13}\boldsymbol{I}_3 = \boldsymbol{V}_1$$

$$R_{21}\boldsymbol{I}_1 + R_{22}\boldsymbol{I}_2 + R_{23}\boldsymbol{I}_3 = \boldsymbol{V}_2$$

$$R_{31}\boldsymbol{I}_1 + R_{32}\boldsymbol{I}_2 + R_{33}\boldsymbol{I}_3 = \boldsymbol{V}_3$$

It is obvious that multiplying both sides of each of these equations by a constant and summing them up does not change the values of the unknown currents $I_1, I_2\ and\ I_3$. Hence, by using this operation repeatedly, in order to eliminate some of the unknowns, it is possible to transform the equation system to the following form:

$$1\boldsymbol{I}_1 + 0\boldsymbol{I}_2 + 0\boldsymbol{I}_3 = C_1$$

$$0\boldsymbol{I}_1 + 1\boldsymbol{I}_2 + 0\boldsymbol{I}_3 = C_2$$

$$0\boldsymbol{I}_1 + 0\boldsymbol{I}_2 + 1\boldsymbol{I}_3 = C_3$$

In which the transformed matrix of coefficients A is the diagonal unitary matrix. It follows directly from these equations that $\boldsymbol{I}_1 = C_1, \boldsymbol{I}_2 = C_2 \; and \; \boldsymbol{I}_3 = C_3$. One of the procedures serving to eliminate some unknowns from the particular equations was demonstrated using Example below.

Example:

Assume that the equation system below, which has the same coefficients, as the system analyzed in the previous Example, i.e;

$$10\mathbf{I}_1 - 7\mathbf{I}_2 = 7$$

$$-3\mathbf{I}_1 + 2\mathbf{I}_2 + 6\mathbf{I}_3 = 4$$

$$5\mathbf{I}_1 - \mathbf{I}_2 + 5\mathbf{I}_3 = 6$$

Using the transformations shown in the previous Example, this equation system can be written in the following equivalent triangular form:

$$10\mathbf{I}_1 - 7\mathbf{I}_2 = 7$$

$$2.5\mathbf{I}_2 + 5\mathbf{I}_3 = 2.5$$

$$6.5\mathbf{I}_3 = 6.2$$

Dividing all three equations by their diagonal coefficients, we obtain:

$$\mathbf{I}_1 - 0.7\mathbf{I}_2 = 0.7$$

$$\mathbf{I}_2 + 2\mathbf{I}_3 = 1$$

$$\mathbf{I}_3 = 1$$

Now we shall eliminate the variable $\mathbf{I}_2$ from the first equation. To do this, we may add to it the second equation multiplied by 0.7. Resulting equation system has the form:

$$\mathbf{I}_1 + 0\mathbf{I}_2 + 1.4\mathbf{I}_3 = 1.4$$

$$\mathbf{I}_2 + 2\mathbf{I}_3 = 1$$

$$\mathbf{I}_3 = 1$$

Next we shall eliminate the variable $\mathbf{I}_2$ from the first and the second equation. It can be done by multiplying the third equation by the constant -1.4 and adding it to the first equation. Similarly, the third equation should be multiplied by -2 and added to the second equation. Finally, we obtain the following system of equations:

$$\mathbf{I}_1 + 0\mathbf{I}_2 + 0\mathbf{I}_3 = 0$$

$$\mathbf{I}_2 + 0\mathbf{I}_3 = -1$$

$$\mathbf{I}_3 = 1$$

For which the matrix of coefficients **A** is diagonal and unitary and has the solution $\mathbf{I}_1 = 0, \mathbf{I}_2 = -1 \; and \; \mathbf{I}_3 = 1$. During the transformation of the equation system from general matrix form (Mesh equation above) to the form of identity matrix, the vector of currents **I** remains unchanged and the operation is made with respect to the matrix of coefficients and vector of voltages. Creation of the so-called augmented matrix of order $n \times (n + 1)$ therefore proves very useful. In case of the system of three equations ($n = 3$) discussed in this section, the **augmented** matrix has the form:

$$\mathbf{R} = [R{:}V] = \begin{bmatrix} R_{11} & R_{12} & R_{13} & V_1 \\ R_{21} & R_{22} & R_{23} & V_2 \\ R_{31} & R_{32} & R_{33} & I_3 \end{bmatrix}$$

Matrix R may be transformed through the full elimination process, after which it takes the form:

$$\begin{bmatrix} 1 & 0 & 0 & I_1 \\ 0 & 1 & 0 & I_2 \\ 0 & 0 & 1 & I_3 \end{bmatrix}$$

For this purpose, the computation algorithm given in the literature can be used.

The LU Matrix Decomposition Method

Let us now consider the task of solving repeatedly the system of linear equations:

$$\mathbf{A} \,.\, \mathbf{X} = \mathbf{B}$$

Each time for the same matrix of coefficients **A**, but for different excitation vectors **B**. The Gauss and Gauss–Jordan elimination methods are not effective for solving this particular problem, because the repeated transformation of the matrix **A** and vector **B** is needed even though the matrix **A** remains the same always. In such case, one of the **LU** decomposition methods, as for example the Crout method, may prove to be more convenient. In this last method, decomposition of the nonsingular matrix **A** of the order n into the product **A = L · U** of the two triangular matrices (lower **L** and upper **U**) is used. Structures of the two matrices **L** and **U** of this product are described by the following general relation:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n-1,1} & a_{n-1,2} & a_{n-1,3} & \cdots & a_{n-1,n} \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} =$$

$$= \begin{bmatrix} l_{11} & 0 & 0 & \cdots & 0 \\ l_{21} & l_{22} & 0 & \cdots & 0 \\ l_{31} & l_{32} & l_{33} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ l_{n-1,1} & l_{n-1,2} & l_{n-1,3} & \cdots & 0 \\ l_{n1} & l_{n2} & l_{n3} & \cdots & l_{nn} \end{bmatrix} \cdot \begin{bmatrix} 1 & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & 1 & u_{23} & \cdots & u_{2n} \\ 0 & 0 & 1 & \cdots & u_{3n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & u_{n-1,n} \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

The equations of the system ($\mathbf{A} \,.\, \mathbf{X} = \mathbf{B}$), which we want to solve, should be written in such an order that the diagonal elements a_{ii} of the coefficient matrix **A** are different from zero and possibly have the greatest absolute values. Then the diagonal elements l_{ii} of the lower triangular matrix **L** will also be different from zero. Substituting the relation **A = L · U** in equations of the system, we obtain:

$$\mathbf{L} \cdot \mathbf{U} \cdot \mathbf{X} = \mathbf{B}$$

Assume initially that the triangular matrices **L** and **U** are known. In consequence, solving the equation system with respect to the column vector **X** can be performed in the two simple stages.

- In the first stage, from the equation:

$$\mathbf{L} \cdot \mathbf{D} = \mathbf{B}$$

We determine the vector D. According to Equation $\mathbf{L} \cdot \mathbf{U} \cdot \mathbf{X} = \mathbf{B}$, this vector satisfies the equation:

$$\mathbf{U} \cdot \mathbf{X} = \mathbf{D}$$

Involving also the desired solution.

- The second stage therefore consists in determining the vector X from Equation $\mathbf{U} \cdot \mathbf{X} = \mathbf{D}$. Both stages of the solution process mentioned above can be performed in a relatively simple way, thanks to the triangular form of the **L** and **U** matrices.

For example, in case of three equations, the system $\mathbf{L} \cdot \mathbf{D} = \mathbf{B}$ takes the form:

$$l_{11}d_1 = b_1$$

$$l_{21}d_1 + l_{22}d_2 = b_2$$

$$l_{31}d_1 + l_{32}d_2 + l_{33}d_3 = b_3$$

And its solution with respect to the vector $\mathbf{D} \equiv [d_1, d_2, d_3]^{\mathrm{T}}$ may be obtained without serious difficulties. In the general case (n > 3), the components d_i of the auxiliary vector **D** can be found by using the following recursive formula:

$$d_1 = \frac{b_1}{l_{11}}$$

$$d_k = \frac{1}{l_{kk}} \cdot \left[b_k - \sum_{i=1}^{k-1} l_{ki} \cdot d_i \right], \qquad k = 2, 3, \dots, n$$

When the column vector **D** is known, we can solve the matrix equation $\mathbf{U} \cdot \mathbf{X} = \mathbf{D}$. Which for n = 3 takes the following form:

$$1x_1 + u_{12}x_2 + u_{13}x_3 = d_1$$

$$1x_2 + u_{23}x_3 = d_2$$

$$1x_3 = d_3$$

The solution $\mathbf{X} \equiv [x_1, x_2,\ x_3]^{\mathrm{T}}$ of these equations can be found in a similarly uncomplicated way, i.e., using the method of consecutive substitutions.

For an arbitrary n > 3, the matrix equation (1.24) has the form:

$$\begin{bmatrix} 1 & u_{12} & u_{13} & \cdots & u_{1n} \\ 0 & 1 & u_{23} & \cdots & u_{2n} \\ 0 & 0 & 1 & \cdots & u_{3n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & u_{n-1,n} \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{n-1} \\ d_n \end{bmatrix}$$

In order to find the solution vector $\mathbf{X} \equiv [x_1, x_2,\ x_3, \ldots, x_n]^{\mathrm{T}}$, the method of consecutive substitutions should be applied. It is defined this time by the following recursive computation formula:

$$x_j = d_j - \sum_{k=j+1}^{n} u_{jk}.x_k \qquad\qquad j = n, n-1, n-2, \ldots, 1$$

According to Equations $\mathbf{L} \cdot \mathbf{D} = \mathbf{B}$ and $\mathbf{U} \cdot \mathbf{X} = \mathbf{D}$, after substituting the new vector **B**, we need to determine only the new vector **D** and next we must calculate the vector **X**, which is the desired solution to our problem. The matrices **L** and **U** need not be reprocessed and this fact diminishes essentially the amount of calculations. It is due to the fact that these matrices were assumed to be known. In the general case, they can be determined using the following recursive relations:

$$l_{i1} = a_{i1}$$

$$l_{ij} = a_{ij} - \sum_{k=1}^{j-1} l_{ik}.u_{kj} \qquad\qquad for\ i \geq j > 1$$

$$u_{1j} = \frac{a_{1j}}{l_{11}}$$

$$u_{ij} = \frac{1}{l_{jj}} \cdot \left[a_{ij} - \sum_{k=1}^{i-1} l_{ik} \cdot u_{kj} \right], \qquad for\ 1 < i < j$$

Which are in the literature often referred to as the Doolittle formulas. The term a_{ij}, where $1 \leq i \leq n, 1 \leq j \leq n$, appearing in these relations, is the element of a given nonsingular matrix of coefficients **A**. These relations are developed in Appendix D of text book Fundamental Numerical Methods for Electrical Engineering, using the standard rule of multiplying two matrices of the same order.

Example: Consider the following system of equations

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 11 & 20 & 29 \\ 3 & 8 & 16 & 24 \\ 4 & 14 & 25 & 40 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4.0 \\ 20.6 \\ 17.4 \\ 27.8 \end{bmatrix}$$

The triangular matrices **L** and **U** determined using the relations (1.30) are equal to:

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 7 & 0 & 0 \\ 3 & 2 & 3 & 0 \\ 4 & 6 & 1 & 4 \end{bmatrix} \qquad \mathbf{U} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The determinant of the matrix of coefficients satisfies the equation:

$$det\mathbf{A} = det\mathbf{L} \cdot det\mathbf{U} = 84 * 1 = 84$$

The solution obtained by using the **LU** decomposition method is: $x_1 = 1.0, x_2 = 0.7, x_3 = 0.4\ and\ x_4 = 0.1$.

The Method of Inverse Matrix

The method of inverse matrix also finds an application for the task of repeatedly solving the system of linear equations:

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$$

For which the matrix of coefficients **A** remains unchanged. In other words, the equation system is being solved for different values of the free terms forming the vector **B**. As we know from the extensive literature on the subject, application of the above method is legitimate, if the number of solution processes applied to the equation system given above is greater than $2n$, where n is the rank of the matrix **A**, equal to the number of equations in the system. The inverse $\mathbf{A^{-1}}$ of a nonsingular square matrix **A** (having the determinant D different from zero) is also the nonsingular square matrix of the same rank. Product of these matrices, i.e.,

$$\mathbf{A^{-1}} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A^{-1}} = \boldsymbol{E}$$

is equal to the unitary matrix **E**, having also the same rank. The equation system will remain unchanged after multiplication of both sides by an inverse matrix, i.e.,

$$\mathbf{A^{-1}} \cdot \mathbf{A} \cdot \mathbf{X} = \mathbf{A^{-1}} \cdot \mathbf{B}$$

Substituting relation $\mathbf{A^{-1}} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A^{-1}} = \boldsymbol{E}$ in the expression above, we obtain:

$$\mathbf{E} \cdot \mathbf{X} = \mathbf{A^{-1}} \cdot \mathbf{B}$$

The product of the unitary matrix **E** of the rank n by a column vector **X** with n elements is identical to the vector **X**. Due to this property, $\mathbf{E} \cdot \mathbf{X} = \mathbf{A^{-1}} \cdot \mathbf{B}$ can be written as:

$$\mathbf{X} = \mathbf{A^{-1}} \cdot \mathbf{B}$$

expressing the essence of the method under consideration. It follows from the above equation that the solution vector **X** can be found by simple multiplication of the inverse matrix $\mathbf{A}^{-1}$ by the vector of free terms **B**. Determination of the inverse matrix $\mathbf{A}^{-1}$ therefore constitutes an essential and most difficult problem, which must be solved in the first stage. Different algorithms available in the literature on linear algebra can be used for this purpose. In case of the matrix of a small rank (n ≤ 3) the relations given in Appendix B of the text book "Fundamental Numerical Methods for Electrical Engineering" may prove to be useful. One of the most popular algorithms used for calculating the inverse matrix is presented below. Assume that a square nonsingular matrix **A** is given. Denote the elements of this matrix by a_{ij}, where $1 \leq i \leq n, 1 \leq j \leq n$. Elements (terms) of the inverse matrix $\mathbf{A}^{-1}$ are denoted by x_{ij}, where $1 \leq i \leq n, 1 \leq j \leq n$. Product of this two matrices, i.e.,

$$\mathbf{A} \cdot \mathbf{A}^{-1} = \boldsymbol{E}$$

Can be presented in the following equivalent form:

$$\sum_{k=1}^{n} a_{ik} x_{kj} = \delta_{ij}$$

Where δ_{ij} is the Kronecker symbol taking the value $1\ for\ i = j$ and the value $0\ for\ i \neq j$. It follows from the above equation that, if we want to determine elements of the column j of the matrix $\mathbf{A}^{-1}$, the following system of equations should be solved:

$$a_{11}x_{1j} + a_{12}x_{2j} + \cdots + a_{1n}x_{nj} = 0$$

$$a_{21}x_{1j} + a_{22}x_{2j} + \cdots + a_{2n}x_{nj} = 0$$

… … … … … … … … … … … … … … … … …

$$a_{j1}x_{1j} + a_{j2}x_{2j} + \cdots + a_{jn}x_{nj} = 1$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$a_{n1}x_{1j} + a_{n2}x_{2j} + \cdots + a_{nn}x_{nj} = 0$$

In order to find all elements of the matrix $\mathbf{A^{-1}}$, the equation system given above should be solved n times, namely for j = 1, 2, 3, ... , n. The matrix of coefficients **A** of this system remains unchanged, and therefore, it can be effectively solved by using the **LU** decomposition method described in the previous section. The product

$$\mathbf{A^{-1}} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A^{-1}} = \boldsymbol{E}$$

Can be used to evaluate precision obtained for the calculated inverse matrix $\mathbf{A^{-1}}$. In case this precision is not satisfactory, the main equation system can be solved once more, this time by using the relation (1.35).

$$\mathbf{X} = \mathbf{A^{-1}} \cdot \mathbf{B}$$

Example: Solve the following system of equations using the method of inverse matrix

$$\begin{bmatrix} 1 & -2 & 3 \\ -1 & 1 & 2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \\ 4 \end{bmatrix}$$

The inverse $\mathbf{A^{-1}}$ of the coefficients matrix **A** of the system given below is equal to (see Appendix B of the text book; Reference [1])

$$\mathbf{A^{-1}} = \frac{1}{8}\begin{bmatrix} -1 & 5 & 7 \\ -3 & 7 & 5 \\ 1 & 3 & 1 \end{bmatrix}$$

According to the relation $\mathbf{X} = \mathbf{A^{-1}} \cdot \mathbf{B}$, we have:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -1 & 5 & 7 \\ -3 & 7 & 5 \\ 1 & 3 & 1 \end{bmatrix}\begin{bmatrix} 12 \\ 8 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \\ 5 \end{bmatrix}$$

Finally, we find our solution: $x_1 = 7, x_2 = 5\ and\ x_3 = 5$.

Indirect or Iterative Methods